

AVNISH SINGH

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SUMMARY

Final-year CS student (graduating May 2027) who builds ML and LLM systems with the checks that make them trustworthy: a coding-agent benchmark whose headline result went against my own thesis, a crisis-planning platform where a solver allocates and deterministic verifiers gate every plan, and a skin-lesion classifier rebuilt after external validation exposed a 7-point drop.

PROJECTS

OmitBench: Benchmark for Silent Omissions in Coding-Agent Patches

2026

Python, AST analysis, mutation testing, LLM-as-judge evaluation, bootstrap CIs, GitHub Actions | github.com/Avnish1505/Omitbench | [engineering blog](#)

- Built a labelled corpus by mutating merged commits from 8 Python libraries into three omission classes: 310 instances, 2,184 requirements per detector, 27.9% base rate, with a CI canary that fails if any detector sees the gold patch.
- The headline result went against my thesis: the AST detector beats a no-AST grep baseline ($\Delta\text{MCC} +0.128$) and a reasoning LLM judge ($+0.111$) but **loses to a mid-tier judge, -0.202** (paired bootstrap CI excludes zero). MCC was pre-committed: flag-everything scores F1 0.44, MCC 0.000.
- Swapping oracle requirements for LLM-extracted ones (a \$0.06 run) dropped MCC from $+0.499$ to -0.935 , which put the bottleneck in extraction, not detection. A pre-registered test of TypeSafe's Jev judge found strong ranking (AUROC 0.936) but failed calibration (ECE 0.323).
- Shipped the AST detector as a GitHub Action PR gate (p95 26.5 ms, 155 tests in CI) because it meets the 0.80 precision bar at $2.6\times$ lower FPR than the winning judge; a gate that cries wolf gets switched off. Two detectors that missed a pre-agreed bar were retired.

AegisOps AI: Human-Approved Crisis Response Planning

2026

Python, OR-Tools CP-SAT, OSRM, LLMs, FastAPI, PostGIS, OpenTelemetry, React, Docker | github.com/Avnish1505/aegisops-ai | [engineering blog](#)

- Replaced LLM-generated resource allocation with a CP-SAT solver over OSRM road travel times on real Lucknow OpenStreetMap data (270 hospitals, 27 police and 3 fire stations); infeasible operator constraints are named, not silently dropped.
- Every plan passes 19 pure-function checks before a person sees it, and fault injection **caught 650/650 faults** across 13 classes with 0 of 50 clean plans falsely blocked. The LLM only reads English, Hindi and Hinglish reports under a JSON schema; unquotable fields are dropped.
- A proposer cannot approve their own plan (JWT roles), every decision lands in a hash-chained audit log and replays to the same verdict, and one OpenTelemetry trace spans each decision. 1,173 passing Python tests, plus Playwright and axe accessibility checks in CI.

Cancer Fusion AI: Multimodal Skin Lesion Classifier

2026

PyTorch, timm, EfficientNetV2-S, ResNet50, conformal prediction, Grad-CAM, FastAPI, React | github.com/Avnish1505/cancer-fusion-ai | [engineering blog](#)

- v1 fused ResNet50 features with age, sex and lesion site on HAM10000 (10,015 images, 7 classes, lesion-level split): 0.712 test macro-F1, but balanced accuracy fell from 0.722 to 0.653 on the official ISIC 2018 external test set.
- Rebuilt as v2 (EfficientNetV2-S at 384 px, colour constancy, sqrt class weights, EMA, 8-view TTA): **external balanced accuracy 0.782** ($+0.129$, paired CI $[+0.088, +0.171]$) with no external drop. Fusion beats image-only by $+0.033$, yet the form adds only $+0.008$ at inference.
- Validation-fitted safety layer: malignant flag at 95% target sensitivity, conformal sets (91.5% coverage on ISIC 2018 at a 90% target), Mahalanobis OOD guard. v2 halves benign lesions flagged per cancer caught (1.72 to 0.86). Opt-in Grad-CAM cut CPU latency ~ 100 to ~ 31 ms.

RESEARCH

Detecting Silent Implementation Integrity Failures in AI-Generated Code: A Static Analysis Approach | [preprint](#), [arXiv pending](#) | [project page](#)

- Deterministic AST analyzer for unwired safety logic: **MCC 0.389** (precision 0.556, recall 0.833) on a 15-scenario benchmark, against a string-presence baseline whose MCC is undefined because it predicts one class. All five errors are reported against limitations written before the run.

Emergence of Artificial Intelligence in Law and Legal Technology | [ADG 2026 International Conference](#) | [paper](#)

- Co-authored a peer-reviewed comparison of AI governance in the EU, US and India; wrote the comparative analysis and technical sections.

TECHNICAL SKILLS

LLM / Agents: LLM evaluation, LLM-as-judge, eval harnesses, structured output, grounded extraction, prompt engineering, AI coding agents

ML: PyTorch, timm, scikit-learn, XGBoost, TensorFlow, computer vision, transfer learning, calibration, conformal prediction, OOD detection

Evaluation: MCC, AUROC, ECE, bootstrap CIs, pre-registered experiments, ablations, leakage guards, mutation testing, AST analysis

Engineering: Python, SQL, FastAPI, PostgreSQL/PostGIS, SQLAlchemy, Docker, GitHub Actions, pytest, React, TypeScript, Playwright, Linux, Git, AWS

EDUCATION

B.Tech, Computer Science and Engineering | Babu Banarasi Das University, Lucknow | CGPA 7.39 / 10

2023 – 2027 (expected May 2027)

ADDITIONAL

- 160 LeetCode problems solved (arrays, strings, hash tables, dynamic programming, DFS) | leetcode.com/u/Avnish1506
- Shortlisted for Anthropic's Claude for Open Source Program on code quality.
- First place, college chess tournament 2025 (Chess.com rapid 1720).